

Technical Requirements
For
Midwave Infrared Relay Lenses
For The
Guided Weapons Evaluation Facility

[REDACTED]

3 April 2025

[REDACTED]
Distribution Statement C: Distribution authorized to U.S. Government agencies and their contractors, Critical Technology; 5 Feb 2025. [REDACTED]

Destruction Notice: Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

Revision History		
Rev. No.	Description	Date
1	Initial Release	20250205
2	Revised physical volume (2 a)	20250321
3	Revised physical volume (2 a) and Optical Requirements (Table 1)	20250403

Abbreviations, Acronyms, and Symbols

g	Force per unit mass due to gravity at the Earth's surface
GWEF	Guided Weapon Evaluation Facility
IR	Infrared
mm	Millimeters
MWIR	Midwave Infrared
μm	Micrometers
”	Inch (Inches)

Table of Contents

Abbreviations, Acronyms, and Symbols	iii
Table of Contents	iv
1. Introduction.....	1
2. Physical Requirements.....	1
3. Environmental Requirements.....	1
4. Optical Performance Requirements	2

1. Introduction

- a. This technical requirements document is for two midwave infrared (MWIR) relay lens assemblies lens assemblies to be used at the Guided Weapon Evaluation Facility (GWEF), [REDACTED]. The two custom optical assemblies will be used to develop a GWEF test system and must be drop-in replacements to an existing GWEF test system.
- b. Contractor proposals should identify any specifications that are not technically feasible or have an undue impact on cost. Should this occur, proposals should provide recommended alternatives to the identified specification as well as impacts on overall system performance and cost.

2. Physical Requirements

- a. The physical volume shall not exceed the following volume:
 - i. MWIR relay lens assembly length: 236.67 mm (9.31")
 - ii. MWIR relay lens assembly width & height: (165.10 mm) 6.5" height
- b. Each lens assembly shall include aluminum dust covers on both ends. In addition, the vendor shall engrave on the cover fiducial marking to assist with alignment and proper clocking.
- c. The vendor shall engrave the relay lens type, vendor name and part number on the front and on the body of the assembly visible with the dust covers on both ends.
- d. Mechanical assembly color: flat black.

3. Environmental Requirements

- a. Operating temperature and humidity range: 65 to 75 degrees Fahrenheit and 20% to 50% humidity, non-condensing. The performance requirement in this document shall be met over this operating range.
- b. Storage temperature and humidity range: 40 to 95 degrees Fahrenheit and 5% to 95% humidity, non-condensing. The lens assembly shall be capable to be stored in an area with this range with no degradation of performance.
- c. Operating shock and vibration range: 3 to 10 g environment. The performance requirement in this document shall be met over this operating range.

4. Optical Performance Requirements

- a. The MWIR relay lens optical requirements are given in Table 1 below. The requirements below marked with an asterisk are defined for a system comprised of the assemble and an existing 16.8" diameter OAP (16" clear aperture), 44.9" vertex focal length, 13.5" zonal radius. The combined system shall work as a collimator with an un-obscured pupil.

Table 1. MWIR Relay Lens Requirements

MWIR Relay Lens Optical Requirements	
Spectral Range	1.9-5.5 μm
Transmission (assessed via witness sample reflectivity)	>70%, 1.9-2.5 μm >80%, 3-5 μm
Encircled Energy*	84% (within 46 μm diameter circle)
Relative Illumination* (at the flat focal plane)	$\geq 94\%$ of $\cos^4\theta$, where θ is any off axis angle within the square 8 degree FOV
Exit Pupil Size at SUT*	≥ 76 mm (2.99")
Exit Pupil Offset*	2377.83 mm (93.62") from SUT aperture to OAP section vertex
Entrance Pupil Diameter*	24.576 mm
Entrance Pupil Offset* (from end of mechanical housing)	109.36 mm (4.31")
Effective Focal Length*	176.3 mm
Front Working Distance	848.17 mm (33.39")
Back Working Distance	109.36 mm (4.31")
Total FOV Presented to SUT*	8-degree circular
Grid Distortion*	<3% across the field of view
Vignetting*	No vignetting within the exit pupil diameter
Ghost images and narcissus*	Optimized to minimize using best engineering practices
Scratch/Dig	60-40
Anti-Reflective Coating	All optical elements will have anti-reflective coatings to preclude ghosting and optimize transmission